Assignment 4: Task 2

Sequential Modeling with Recurrent Networks

Spoken Digit Sequential Classification (Pre-computed MFCC Sequences)

1. Dataset and Preprocessing

The dataset contains 8,800 utterances represented by 13-dimensional MFCC feature sequences for spoken digits 0–9. The training set contains 6,600 utterances and the test set contains 2,200 utterances. Sequence lengths range from 4 to 93 frames, with a mean of 39.8 frames.

Dataset split	**Utterances**
Training before validation split	6,600
Training after validation split	5,940
Validation	660
Test	2,200
Feature dimensions	13
Classes	10 (digits 0–9)
Sequence length (min / mean / max)	4 / 39.8 / 93

Sequences were normalized and padded for batching. Variable-length sequences were packed for recurrent processing. A stratified 10% validation split was taken from the original training data.

2. Model Configurations

Vanilla RNN, LSTM, and GRU models were evaluated using 1, 2, or 3 recurrent layers and 32, 64, or 128 hidden units. Training used Adam with learning rate 0.001, gradient clipping at 5.0, and a convergence tolerance of 10^{-4} for the change in average training loss. The safety cap was 150 epochs.

Cell	**Layers**	**Hidden units**	**Trainable parameters**
RNN	1	32	1,834
RNN	1	64	5,706
RNN	1	128	19,594
RNN	2	32	3,946
RNN	2	64	14,026
RNN	2	128	52,618
RNN	3	32	6,058
RNN	3	64	22,346
RNN	3	128	85,642
LSTM	1	32	6,346
LSTM	1	64	20,874
LSTM	1	128	74,506
LSTM	2	32	14,794
LSTM	2	64	54,154
LSTM	2	128	206,602
LSTM	3	32	23,242
LSTM	3	64	87,434
LSTM	3	128	338,698
GRU	1	32	4,842
GRU	1	64	15,818

GRU	1	128	56,202
GRU	2	32	11,178
GRU	2	64	40,778
GRU	2	128	155,274
GRU	3	32	17,514
GRU	3	64	65,738
GRU	3	128	254,346

3. Hyperparameter Results

Cell	Layers	Hidden	Train acc.	Test acc.	Epochs	Parameters
RNN	1	32	97.58%	85.86%	142	1,834
RNN	1	64	97.22%	88.73%	150	5,706
RNN	1	128	98.13%	90.64%	150	19,594
RNN	2	32	97.91%	90.18%	146	3,946
RNN	2	64	99.09%	92.36%	150	14,026
RNN	2	128	99.51%	94.27%	150	52,618
RNN	3	32	94.43%	86.73%	150	6,058
RNN	3	64	99.12%	94.09%	150	22,346
RNN	3	128	98.94%	92.59%	150	85,642
LSTM	1	32	99.98%	94.77%	71	6,346
LSTM	1	64	100.00%	94.73%	39	20,874
LSTM	1	128	100.00%	97.05%	25	74,506
LSTM	2	32	99.83%	95.14%	30	14,794
LSTM	2	64	100.00%	95.77%	29	54,154
LSTM	2	128	100.00%	95.77%	21	206,602
LSTM	3	32	99.55%	95.00%	23	23,242
LSTM	3	64	100.00%	96.36%	34	87,434
LSTM	3	128	100.00%	96.23%	39	338,698
GRU	1	32	99.95%	93.45%	40	4,842
GRU	1	64	100.00%	94.59%	35	15,818
GRU	1	128	100.00%	97.05%	32	56,202
GRU	2	32	99.98%	96.32%	48	11,178
GRU	2	64	100.00%	95.95%	31	40,778
GRU	2	128	100.00%	97.00%	22	155,274
GRU	3	32	100.00%	95.73%	45	17,514
GRU	3	64	100.00%	96.00%	32	65,738
GRU	3	128	100.00%	96.95%	19	254,346

Cell	Mean test accuracy	Best test accuracy	Worst test accuracy	Mean epochs	Mean parameters	Not converged
GRU	95.89%	97.05%	93.45%	33.78	69,077	0
LSTM	95.65%	97.05%	94.73%	34.56	91,850	0
RNN	90.61%	94.27%	85.86%	148.67	23,530	7

The RNN configurations did not satisfy the convergence criterion within the 150-epoch cap in seven cases. All tested LSTM and GRU configurations finished before the cap.

4. Training Loss versus Epochs

Training-loss curves for the explored configurations are included below. RNN runs generally required more epochs; LSTM and GRU runs converged sooner.

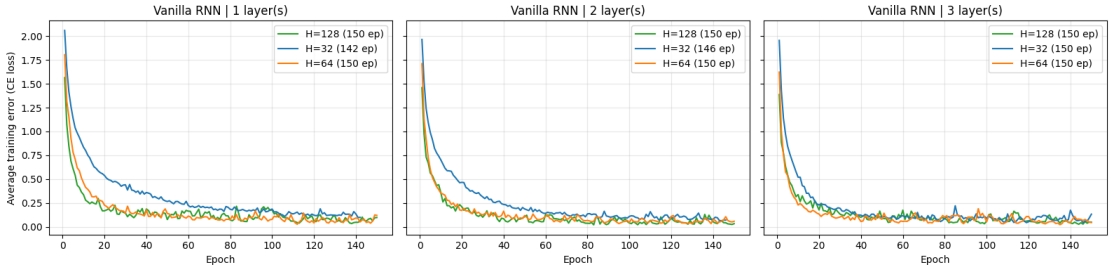


Figure 1. Vanilla RNN loss curves across hidden sizes.

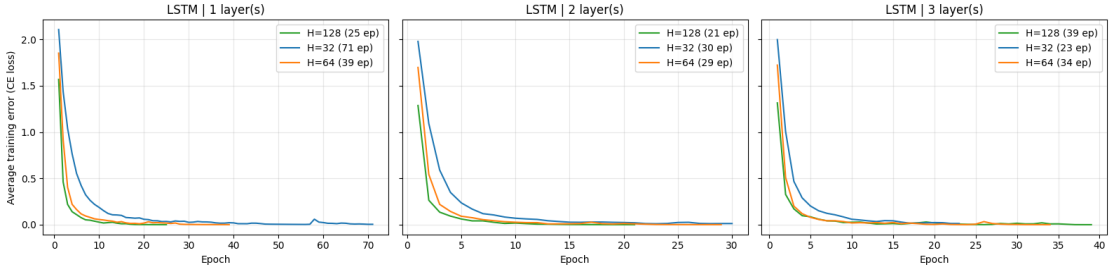


Figure 2. LSTM loss curves across hidden sizes.

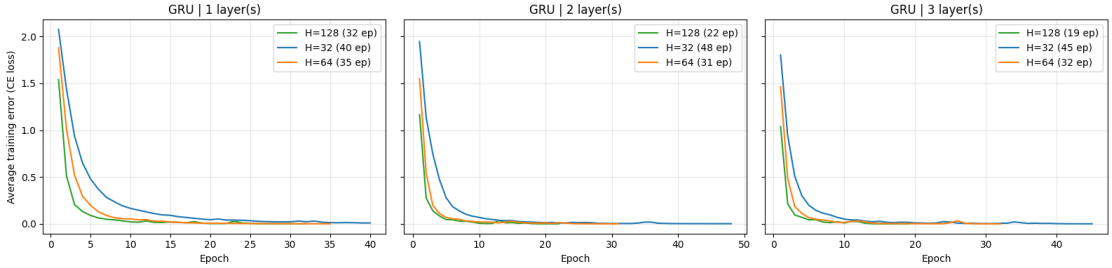


Figure 3. GRU loss curves across hidden sizes.

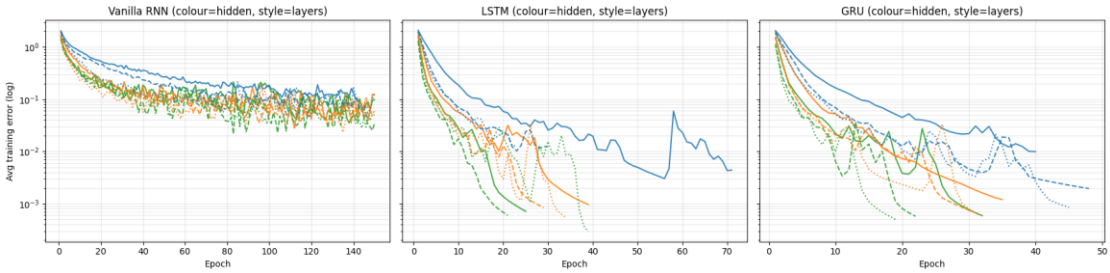


Figure 4. Loss curves across explored configurations.

5. Best Configuration for Each Recurrent Cell

Model	Best configuration	Parameters	Epochs	Train accuracy	Validation accuracy	Test accuracy
Vanilla RNN	2 layers, 128 units	52,618	150	99.51%	97.88%	94.27%
LSTM	2 layers, 64 units	54,154	29	100.00%	99.85%	95.77%
GRU	3 layers, 64 units	65,738	32	100.00%	100.00%	96.00%

GRU + attention (bonus)	3 layers, 64 units	65,803	25	100.00%	99.85%	98.64%

The bonus GRU with attention pooling achieved the highest reported test accuracy of 98.64%.

6. Confusion Matrices

Each test set contains 220 examples per digit class. Rows represent true digit labels and columns represent predicted labels.

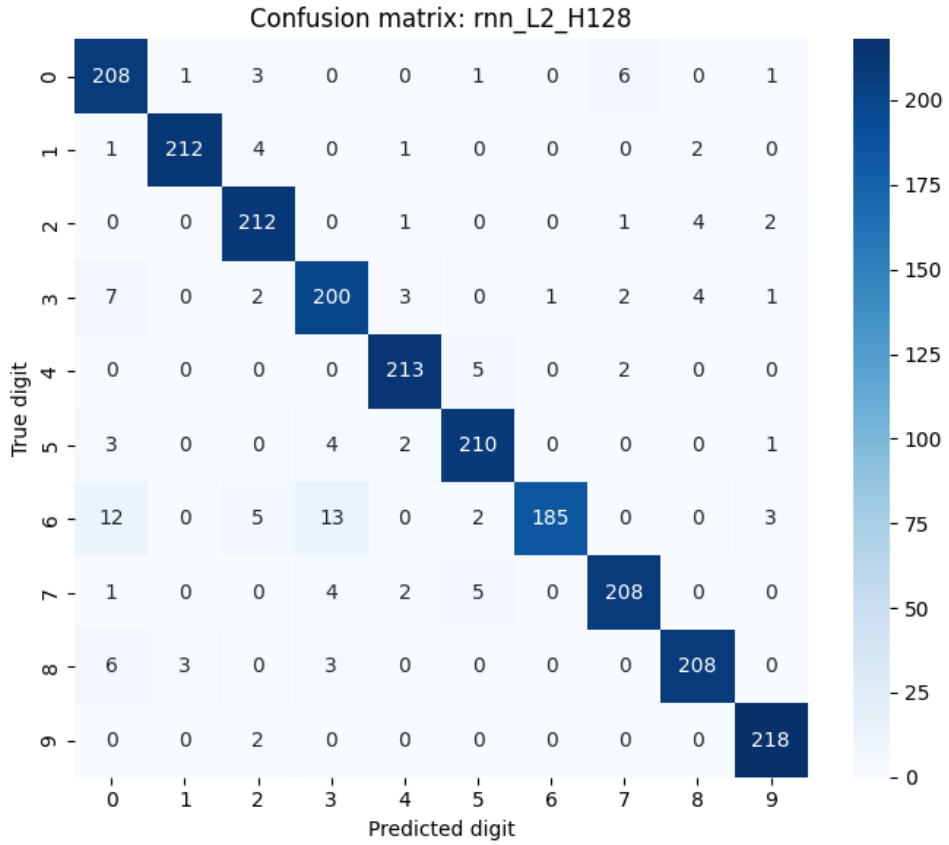


Figure 5. Best Vanilla RNN: rnn_L2_H128 (94.27% test accuracy).

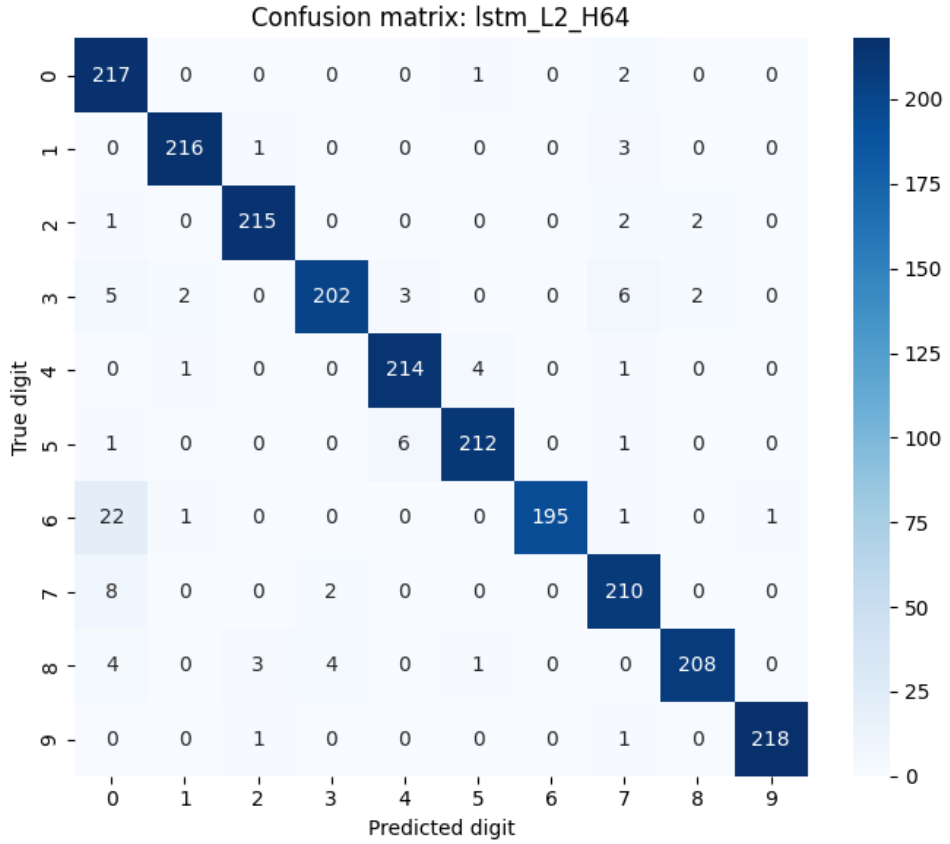


Figure 6. Best LSTM: lstm_L2_H64 (95.77% test accuracy).

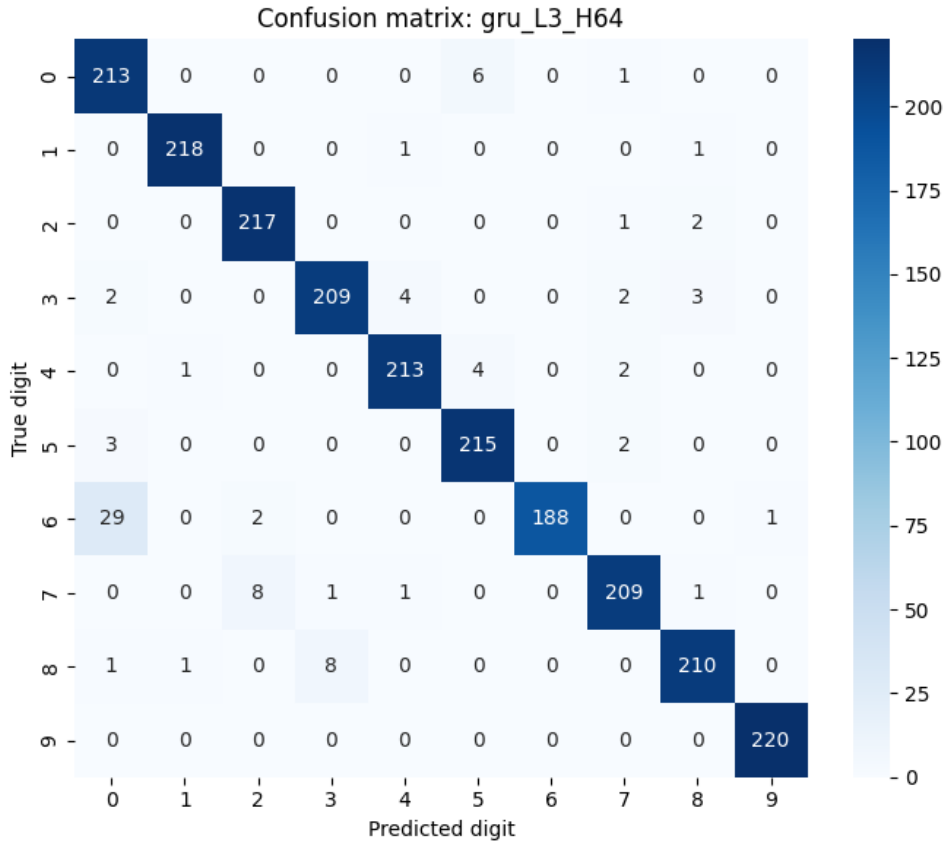


Figure 7. Best GRU: gru_L3_H64 (96.00% test accuracy).

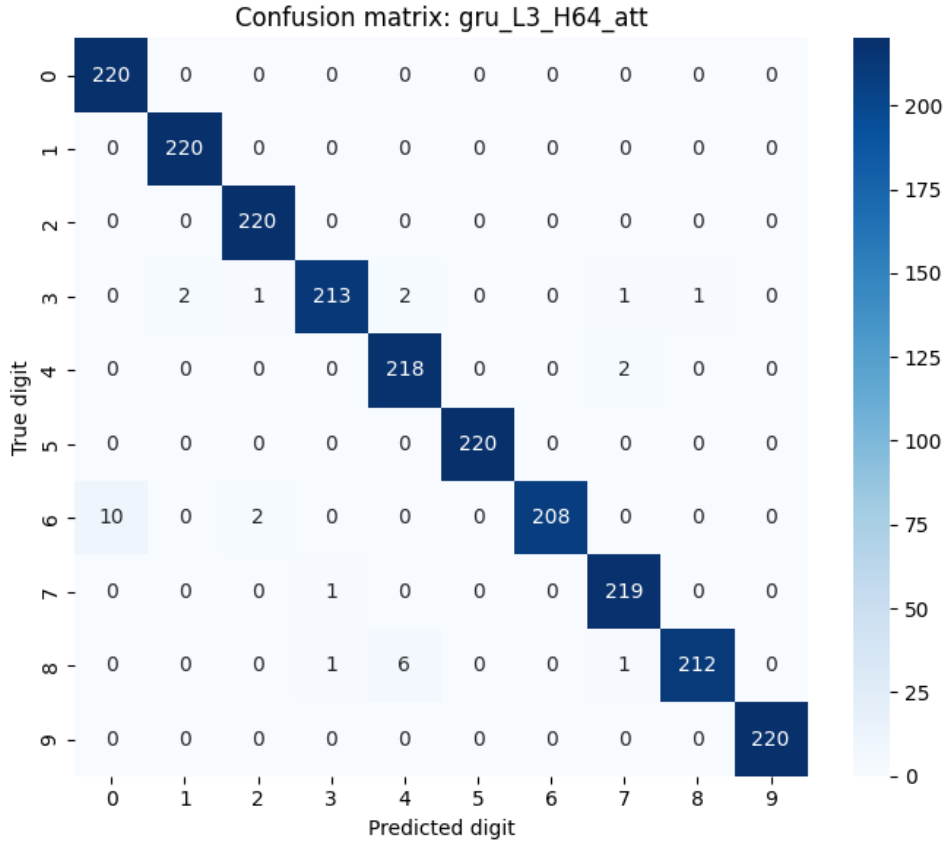


Figure 8. Best bonus model: gru_L3_H64_att (98.64% test accuracy).

7. Detailed Classification Metrics

Digit	RNN F1	LSTM F1	GRU F1	GRU + attention F1	Support
0	0.9083	0.9079	0.9103	0.9778	220
1	0.9725	0.9818	0.9909	0.9955	220
2	0.9464	0.9773	0.9709	0.9932	220
3	0.9009	0.9439	0.9543	0.9793	220
4	0.9638	0.9661	0.9704	0.9776	220
5	0.9481	0.9680	0.9663	1.0000	220
6	0.9113	0.9398	0.9216	0.9720	220
7	0.9476	0.9396	0.9565	0.9887	220
8	0.9498	0.9630	0.9611	0.9792	220
9	0.9776	0.9932	0.9977	1.0000	220
Macro F1	0.9426	0.9581	0.9600	0.9863	2,200

8. MFCC Feature Visualizations

Representative utterance heatmaps show the 13 MFCC coefficients over time for samples from the ten digit classes.

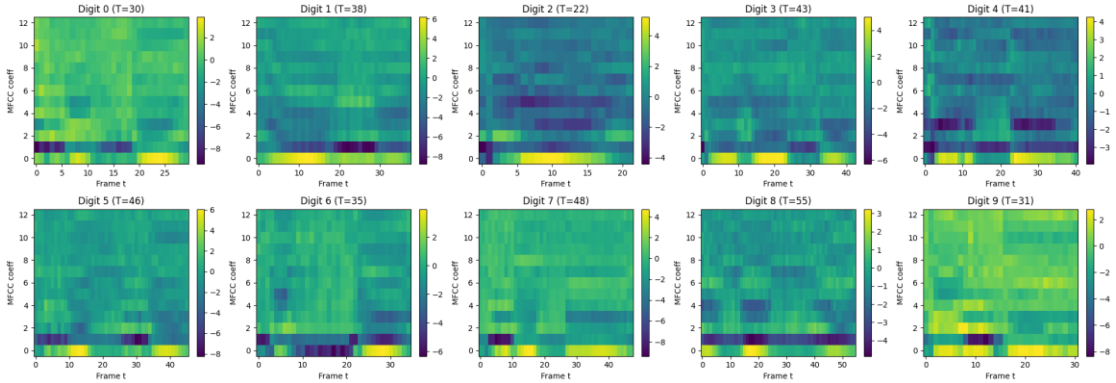


Figure 9. MFCC coefficient heatmaps for one utterance from each digit class.

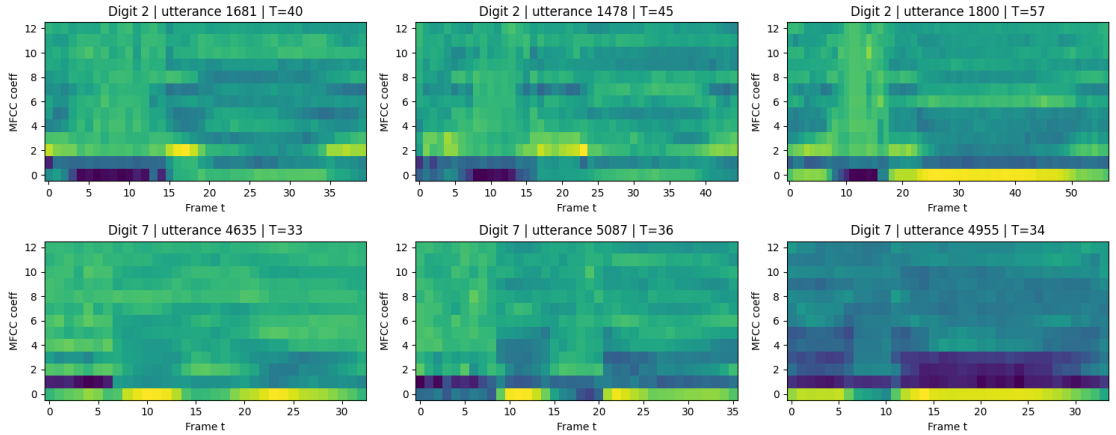


Figure 10. Additional sequential MFCC feature visualization.

9. Gradient-Flow Analysis

The reported ratio compares the gradient norm at the first recurrent layer with that at the top recurrent layer. For one-layer models the ratio is 1.00 by definition.

Cell	Layers	Hidden 32	Hidden 64	Hidden 128
GRU	1	1.000	1.000	1.000
GRU	2	1.933	2.230	2.373
GRU	3	2.695	2.411	2.721
LSTM	1	1.000	1.000	1.000
LSTM	2	0.994	1.370	1.746
LSTM	3	0.849	1.518	2.327
RNN	1	1.000	1.000	1.000
RNN	2	1.389	1.442	1.525
RNN	3	1.196	1.566	1.493

Cell	Layers	Hidden 32	Hidden 64	Hidden 128
GRU	1	0.789	0.426	0.430
GRU	2	0.448	0.393	0.443
GRU	3	0.453	0.324	0.429
LSTM	1	1.171	0.794	0.892
LSTM	2	1.411	0.670	0.541
LSTM	3	1.538	0.638	0.542
RNN	1	12.247	6.559	8.166

RNN	2	8.439	5.109	3.676
RNN	3	7.183	3.552	3.661

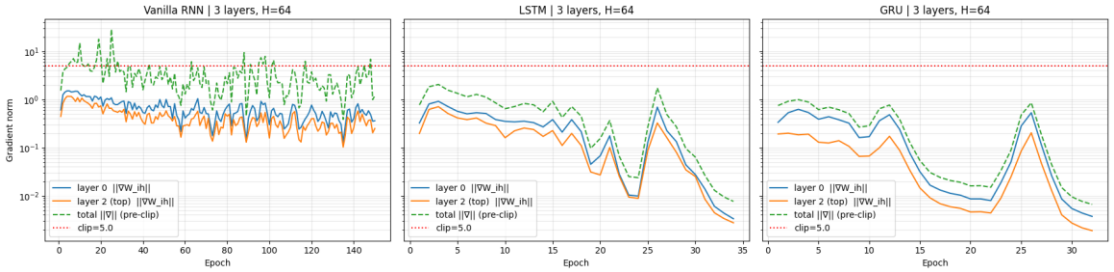


Figure 11. Gradient-flow visualizations.

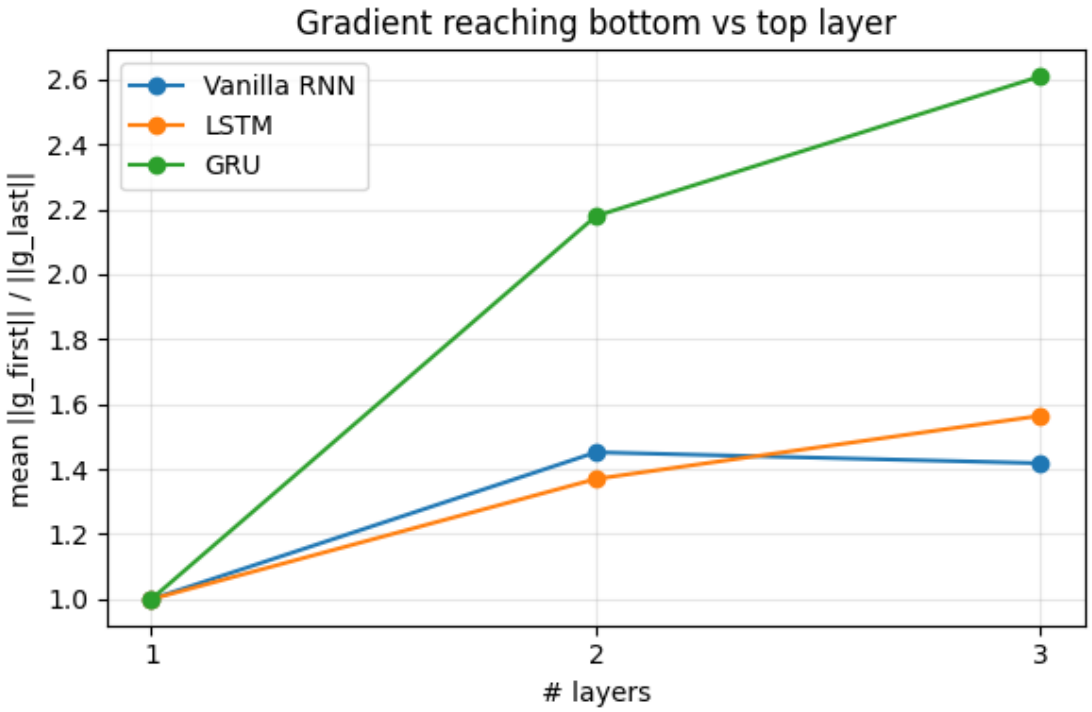


Figure 12. Additional gradient-flow comparison.

10. Bonus Models

Bidirectional GRU, attention pooling, and combined bidirectional-attention variants were evaluated.

Bonus configuration	**Test accuracy**	**Epochs**	**Parameters**
GRU + bidirectional	As reported in notebook	—	—
GRU + attention	98.64%	25	65,803
GRU + bidirectional + attention	As reported in notebook	—	—

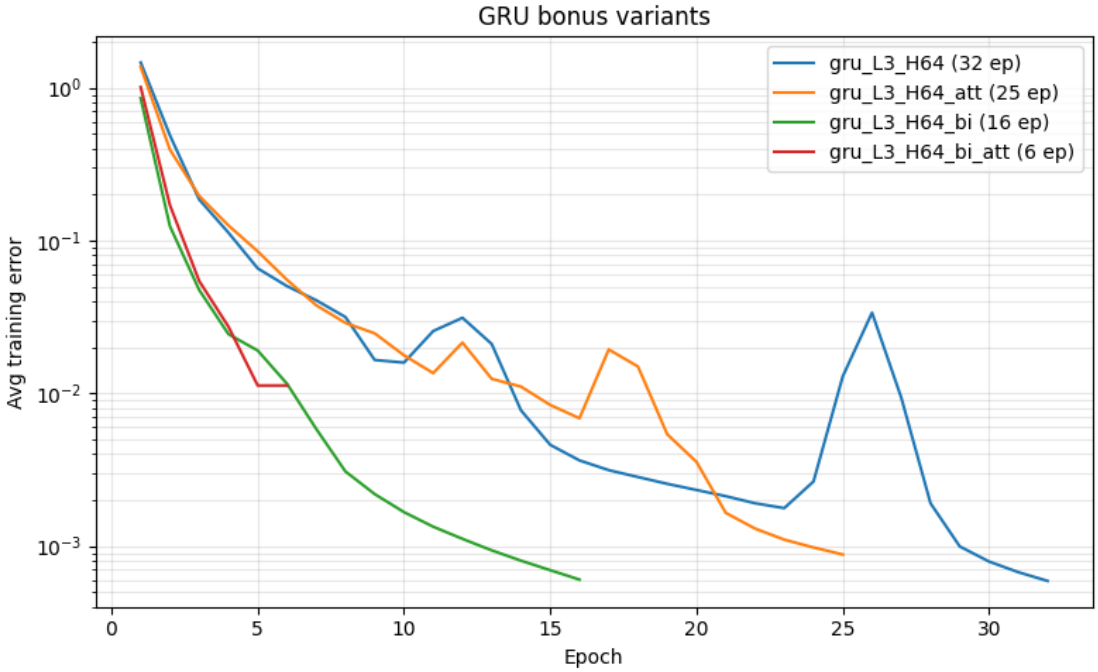


Figure 13. Bonus architecture comparison from the notebook.

11. Key Observations

• The GRU family had the highest mean test accuracy across the standard grid (95.89%), followed by LSTM (95.65%) and Vanilla RNN (90.61%).

• The widest GRU and LSTM configurations reached 97.05% test accuracy in the standard grid.

• Vanilla RNN training was slower, and seven configurations did not meet the convergence criterion within 150 epochs.

• For the best standard configurations, digit 6 was a frequent source of errors.

• Attention pooling improved the reported GRU test accuracy from 96.00% to 98.64%, an increase of 2.64 percentage points.